7\. Database Indexing

Database indexes improve query performance by reducing the amount of data a database must examine when searching for records. Without an appropriate index, a database may perform a full table scan, examining every row to find matching results. B-tree indexes are commonly used for equality comparisons, range queries, and ordered retrieval. Hash indexes are optimized for equality comparisons but generally do not support range queries. In vector databases, approximate nearest neighbor indexes improve the speed of similarity searches by avoiding exhaustive comparisons against every stored vector. One popular approach is Hierarchical Navigable Small World (HNSW), which organizes vectors into multiple graph layers. Although indexes improve retrieval speed, they consume additional storage and introduce overhead during insertions, updates, and deletions.

8\. Microservices Architecture

Microservices architecture is an application design approach in which a system is divided into independently deployable services. Each service is responsible for a specific business capability and communicates with other services through APIs or messaging systems. For example, an e-commerce platform might separate authentication, product catalog management, payment processing, inventory management, and order fulfillment into individual services. This architecture allows teams to develop and deploy services independently and select technologies appropriate for each component. However, microservices increase operational complexity because developers must manage service discovery, distributed tracing, network communication, authentication, and failure handling. A monolithic architecture may be more appropriate for smaller applications that do not require independent scaling or deployment.